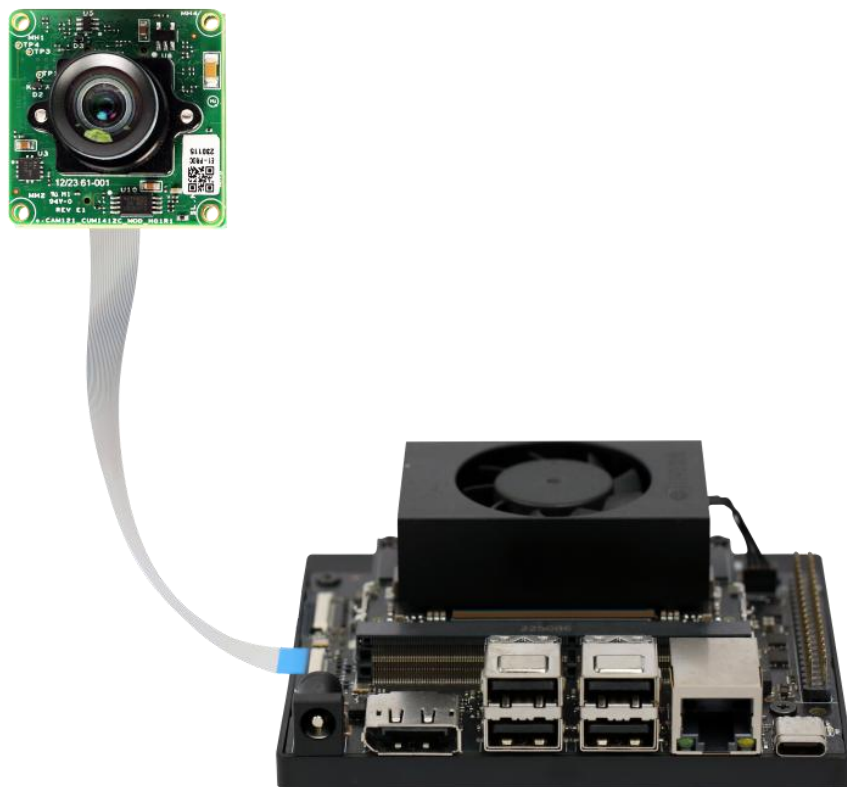


e-CAM121_CUONX

eCAM_Argus_App_User_Manual



Version 1.2

e-con Systems

8/27/2023

Disclaimer

e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.

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Introduction to e-CAM121_CUONX

e-CAM121_CUONX is a 12 MP color camera from e-con Systems, a company with over two decades of experience in designing, developing, and manufacturing OEM cameras. It is interfaced to a J21 connector of a Jetson Orin Nano™/Orin™ NX development kit using an ACC-ONX-MI-WTB-ADP-H01 adaptor board in 4-lane and also provided with a S-mount (also known as M12 board lens) lens holder. The S-mount is one of the most used small form-factor lens mounts for board camera.

e-CAM121_CUONX camera is a MIPI camera and contains an 1/2.3" IMX412-AACK-D CMOS image sensor from SONY®. It requires a custom driver for the NVIDIA® Jetson Orin Nano™/Orin™ NX development kit. e-con Systems provides the prebuilt driver for this camera along with the camera board.

e-con Systems provides a sample application, called eCAM_argus_camera, a customized version of argus_camera sample application provided by NVIDIA®, along with the e-CAM121_CUONX camera. It is a video viewer software customized to demonstrate the features of e-CAM121_CUONX.

This document describes about the usage of application and the special features of application when it is used with e-CAM121_CUONX on Linux operating systems (OS).

Description

This IMX412-AACK-D based camera in e-CAM121_CUONX can stream in 4-lane mode with the NVIDIA® Jetson Orin Nano™/Orin™ NX development kit.

The supported resolutions and frame rates in 4-lane mode are listed in the below table.

Table 1: Supported Resolutions and Frame Rates in 4-Lane Mode

S.No	Resolution	Frame Rate (fps)
1	4056 x 3040 (10 BPP)	60
2	4056 x 3040 (12 BPP)	40
3	4056 x 3040 (12 BPP)	30
4	4056 x 2288 (10 BPP)	60
5	4056 x 2288 (10 BPP)	30
6	2028 x 1112 (10 BPP)	240
7	2028 x 1112 (10 BPP)	60

The camera controls of e-CAM121_CUONX are as follows:

- Exposure Time Range
- Gain Range
- ISP Digital Gain Range
- Sensor Mode Index
- Frame Rate
- Output size
- Output path
- De-Noise Mode
- De-Noise Strength
- Edge Enhance Mode
- Edge Enhance Strength
- Auto Exposure (AE) Antibanding Mode
- Auto White Balance (AWB) Mode
- Exposure Compensation
- AE Lock
- AWB

The eCAM_argus_camera is a sample libArgus based camera application used to demonstrate the functioning of e-CAM121_CUMI412C_MOD_H01R1 module boards in Jetson Orin Nano™/Orin™ NX development kit.

Using eCAM_argus_camera application, you can perform the following:

- Stream all available resolutions if different resolutions are supported by the device.
- Change controls for connected camera.
- Capture images in RAW and JPEG formats.
- Record H264, H265 encoded videos.

e-con Systems provides prebuilt binary of the application along with source code. Please refer to the e-CAM121_CUONX_Release_Package_Manifest_Rev_<ver>.pdf for the compatible Linux distribution version (L4T version).

Launching Application

This section describes how to launch the eCAM_argus_camera application.

The commands and output messages in this manual are represented by different colors as listed in the below table.

Table 2: Notation of Colors

Color	Notation
Blue	Commands running in development kit
Red	Output message in development kit

The steps to launch the application are as follows:

1. Connect the e-CAM121_CUONX camera module to the development kit.
2. Run the following command to check whether the camera is initialized.

```
$ sudo dmesg | grep -i "Detected eimx412 sensor"
```

The output message appears as shown below.

```
Detected eimx412 sensor
```

The output messages indicate that the camera is initialized properly.

3. Run the following command to check the presence of video node.

```
$ ls /dev/video*
```

The output message for camera setup appears as shown below.

```
/dev/video0
```

4. Run the following command to achieve maximum frame rate before launching the eCAM_argus_camera application in the Jetson Orin Nano™/Orin™ NX development kit.

```
$ sudo ~/max-vi-isp-clks.sh
```

5. Run the following command to launch the eCAM_argus_camera application.

```
$ /usr/local/bin/eCAM_argus_camera --device=<n>
```

Where <n> refers to the device node.

Once the application is launched, the control window and video display window appear as shown below.

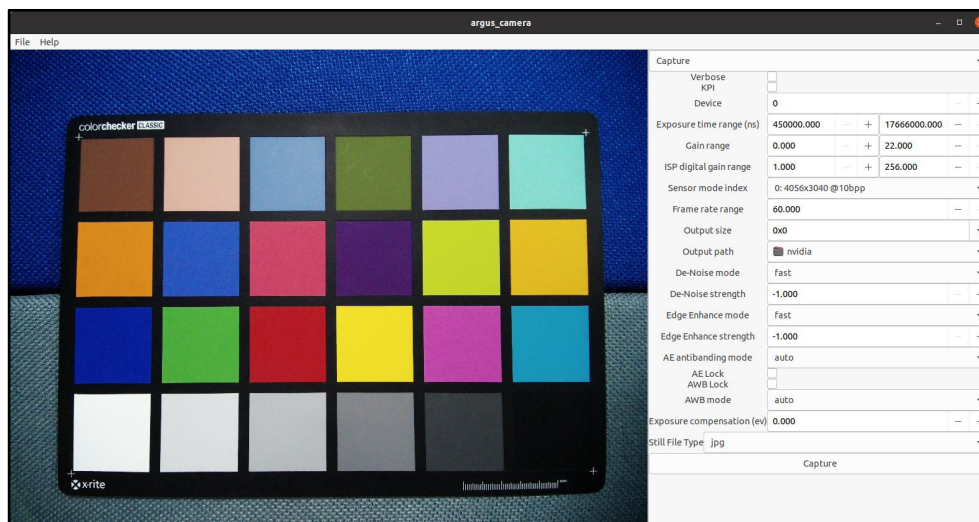


Figure 1: Initial Control Window and Video Display Window Appearance

Application Features

This section describes the controls that are supported in the current version of eCAM_argus_camera application for e-CAM121_CUONX.

The sensor features that are supported in the current version of eCAM_argus_camera application are as follows:

- [Exposure Time Range](#)
- [Gain Range](#)
- [Sensor Mode Index](#)
- [Frame Rate](#)

The camera ISP features supported in the current release version of eCAM_argus_camera application are as follows:

- [ISP Digital Gain Range](#)
- [De-Noise Mode](#)
- [Output Size](#)
- [Output Path](#)
- [De-Noise Strength](#)
- [Edge Enhance Mode](#)
- [Edge Enhance Strength](#)
- [AE Antibanding Mode](#)
- [AWB Mode](#)
- [Exposure Compensation](#)
- [AE Lock](#)

In eCAM_argus_camera application, the sensor and camera ISP features appear as shown below.

Capture					
Verbose KPI	☐				
Device	0				- +
Exposure time range (ns)	450000.000	-	+	17666000.000	- +
Gain range	0.000	-	+	22.000	- +
ISP digital gain range	1.000	-	+	256.000	- +
Sensor mode index	0: 4056x3040 @10bpp				
Frame rate range	60.000				- +
Output size	0x0				▼
Output path	nvidia				
De-Noise mode	fast				
De-Noise strength	1.000				- +
Edge Enhance mode	fast				
Edge Enhance strength	-1.000				- +
AE antibanding mode	auto				
AE Lock	☐				
AWB Lock	☐				
AWB mode	auto				
Exposure compensation (ev)	0.000				- +
Still File Type	jpg				
Capture					

Figure 2: Controls of e-CAM121_CUONX

You can click the respective control box and configure the preview settings according to your needs. The value being set will be displayed in the respective text box. Once the value is set to the respective control box, you can notice the change in preview's property.

The values of e-CAM121_CUONX controls are listed in the below table.

Table 3: Values of e-CAM121_CUONX Controls

Controls	Minimum Value	Maximum Value	Default Value	Manual Control	Auto Control
Exposure time range (ns)	450000	26000000	25000000	Yes	Yes
Gain range (Af)	0	22	0	Yes	Yes
ISP digital gain range	1	256	-	Yes	Yes
Sensor mode index	0	3	0	Yes	No
Frame rate (fps)	1	60	60	Yes	No

Output size	0x0	3840x2160	0x0	Yes	No
De-Noise mode	off	High quality	fast	Yes	No
De-Noise strength	-1.00	1.00	-1.00	Yes	No
Edge enhance mode	off	High quality	fast	Yes	No
Edge enhance strength	-1.00	1.00	-1.00	Yes	No
AE antibanding mode	Off	60hz	auto	Yes	No
AWB mode	auto	Manual	auto	Yes	No
Exposure compensation (eV)	-2	+2	0	Yes	No

Note: The application currently does not support focus position control since the camera has a custom lens which uses only manual focus.

Exposure Time Range

The exposure time range can be changed from a minimum value of 450000 to a maximum value of 26000000 by entering the values or selecting the value in the **Exposure time range (ns)** box. The exposure time is represented in Nanoseconds (ns) scale. The default exposure time range set by the application depends upon the luminance of the source. However, to set a specific exposure time range value, you can manually set both the lower and higher range fields to that same specific value in the **Exposure time range (ns)** box as shown below.

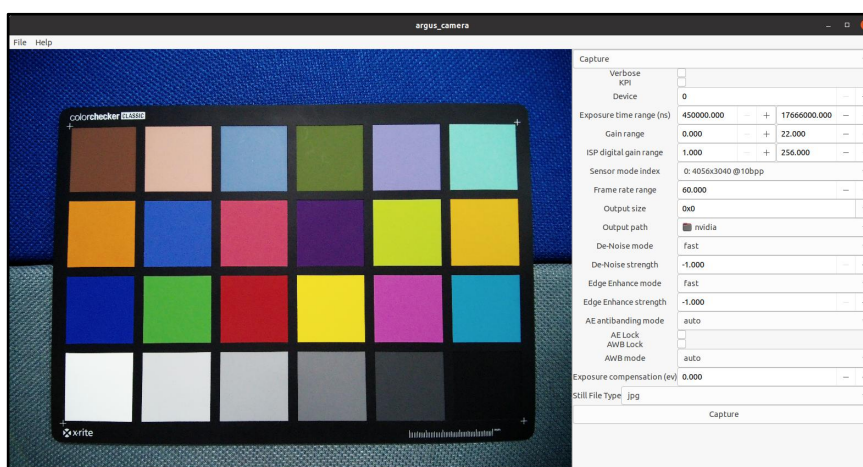


Figure 3: Exposure Time Range

Note: The exposure control is limited to the frame rate as the application is designed to achieve a constant frame rate. For example, the maximum exposure that can be achieved is limited to maximum of 10 ms even a larger exposure range value is specified in the **Exposure time range (ns)** box, when the frame rate control is set to different fps.

Gain Range

The gain range which corresponds to sensor gain range can be changed from a minimum value of 1 to a maximum value of 22. The gain range value is represented in amplification factor (A_f). IMX412-AACK-D can support a sensor gain range of 0 to 27 dB. The default gain value set depends upon the luminance of the source. However, to set a specific constant sensor gain value, you can manually set both the lower and higher range fields to that same specific value in the **Gain range** box as shown below.

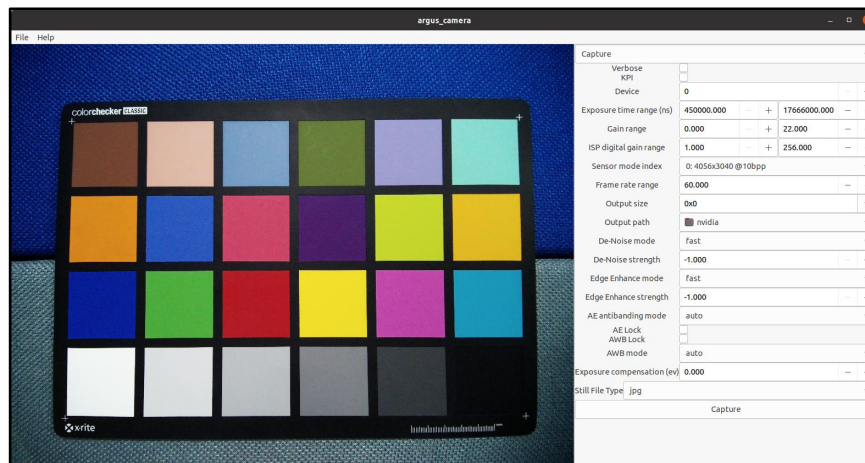


Figure 4: Gain Range

Note: The sensor gain range and exposure time range changes are local. On changing the modes, the corresponding values will be discarded, and the default range values will be applied.

Sensor Mode Index

The sensor mode index contains all the combinations of resolutions, frame rate and bpp (bits per pixel) that the camera sensor outputs.

The sensor modes currently supported by the e-CAM121_CUONX are listed in the below table.

Table 4: Modes and Resolutions Supported in 4-Lane

0	4056 x 3040 (10 BPP)	60
1	4056 x 3040 (12 BPP)	40

2	4056 x 3040 (12 BPP)	30
3	4056 x 2288 (10 BPP)	60
4	4056 x 2288 (10 BPP)	30
5	2028 x 1112 (10 BPP)	240
6	2028 x 1112 (10 BPP)	60

Note: The above sensor modes are the direct inputs from the camera sensor.

Frame Rate

The **Frame rate range** box shows the maximum frames per second for that particular mode. You can view the screen similar to the screen shown below.

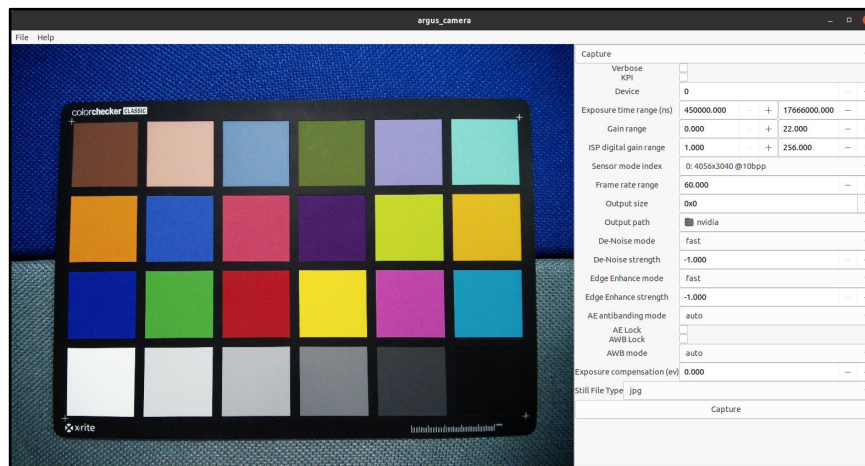


Figure 5: Frame Rate

Note: Frame rate displayed (when KPI is enabled) is the number of frames from streaming camera per second. Also, the FPS displayed will not go beyond the refresh rate of the monitor used.

ISP Digital Gain Range

The ISP gain control can be changed from a minimum value of 1 to a maximum of 256. This gain corresponds to the digital gain additionally provided by the ISP (over and above the sensor gain) and the default value depends upon the luminance of the source. Jetson Orin Nano™/Orin™ NX ISP can support a digital gain range of 1 to 256 (A_f) corresponding to 0 to 36.16 dB. However, in e-CAM121_CUONX, ISP gain has been restricted to 1 - 256 (A_f), including the image quality and other performance metrics. To manually set the ISP digital gain, you can change the lower range value and the higher range value to a constant in the **ISP digital gain range** box as shown below.

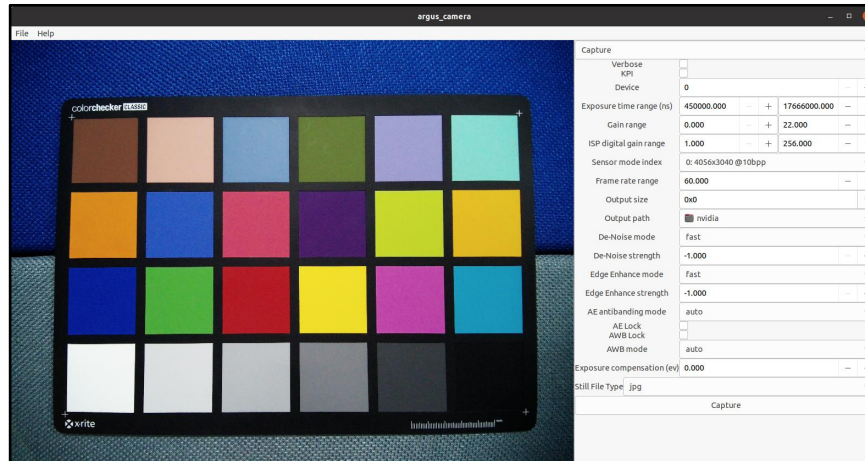


Figure 6: ISP Digital Gain Range

De-Noise Mode

The eCAM_argus_camera application provides two types of de-noise modes as follows:

- Fast
- High Quality

You can select the required de-noise mode from the **De-Noise mode** drop-down list box. The default de-noise mode is fast. When you click the **De-Noise mode** drop-down list box, the supported de-noise modes will be listed as shown below.

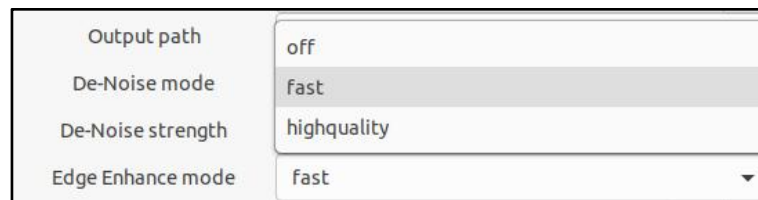


Figure 7: De-Noise Mode

Output Size

The eCAM_argus_camera application provides four types of output frame sizes from the centre of the image.

You can select the required **Output size mode** from the **Output size mode** drop-down list box. The default output size mode is 0x0 which gives the maximum resolution supported by the sensor i.e., 4056x3040. When you click the **Output size mode** drop-down list box, the supported output size mode will be listed as shown below.

Output path	0x0
De-Noise mode	176x144
De-Noise strength	320x240
Edge Enhance mode	640x480
Edge Enhance strength	1280x720
AE antibanding mode	1920x1080
AE Lock	3840x2160

Figure 8: Output Size

Output Path

The output path denotes the path (directory) in which you want to store your captured files.

When you click the **Output Path** drop-down list box, the directories will be listed as shown below.

Output path	nvidia
De-Noise mode	Desktop
De-Noise strength	File System
Edge Enhance mode	L4T-README
Edge Enhance strength	Documents
AE antibanding mode	Music
AE Lock	Pictures
AWB Lock	Videos
AWB mode	Downloads
Exposure compensation (ev)	Other...

Figure 9: Output Path

De-Noise Strength

You can change the de-noise strength value from a minimum value of -1.0 to a maximum value of 1.0. The default de-noise strength value is -1.0 and can be changed by entering the values or selecting the values in the **De-Noise strength** box. You can view the screen similar to the screen shown below.

Output path	 nvidia	▼
De-Noise mode	fast	
De-Noise strength	0.200	– +
Edge Enhance mode	fast	
Edge Enhance strength	-1.000	– +

Figure 10: De-Noise Strength

Note: When the de-noise mode is set to high quality, there will be no significant change in de-noising in this L4T release (will be upgraded later).

Edge Enhance Mode

The eCAM_argus_camera application provides two types of edge enhance modes as follows:

- Fast
- High Quality

You can select the required edge enhancement mode from the **Edge Enhance mode** drop-down list box. The default edge-enhance mode is fast. When you click the **Edge Enhance mode** drop-down list box, the supported edge enhancement modes will be listed as shown below.

De-Noise mode	fast	▼
De-Noise strength	off	
Edge Enhance mode	fast	
Edge Enhance strength	highquality	
AE antibanding mode	auto	

Figure 11: Edge Enhance Mode

Edge Enhance Strength

You can change the edge enhance strength value from a minimum value of -1.0 to a maximum value of 1.0. The default edge enhance strength value is -1.0 and can be changed by manually inserting the values or selecting the values in the **Edge Enhance strength** drop-down list box. You can view the screen similar to the screen shown below.

Output path	nvidia	
De-Noise mode	fast	
De-Noise strength	-1.000	- +
Edge Enhance mode	fast	
Edge Enhance strength	-1.000	- +

Figure 12: Edge Enhance Strength

AE Antibanding Mode

The eCAM_argus_camera application supports the AE antibanding modes as follows:

- Off
- Auto
- 50 Hz
- 60 Hz

By default, the AE antibanding mode is auto. You can select the required AE antibanding mode from the **AE antibanding mode** drop-down list box. When you click the **AE antibanding mode** drop-down list box, the list of AE antibanding modes appear as shown below.

De-Noise strength	1.000	- +
Edge Enhance mode	fast	
Edge Enhance strength	off	
AE antibanding mode	auto	
AE Lock		
AWB Lock	50hz	
AWB mode	60hz	
Exposure compensation (ev)	0.000	- +

Figure 13: AE Antibanding Mode

AWB Mode

The eCAM_argus_camera application automatically detects the scene temperature and adjusts the white balance. The supported AWB modes are as follows:

- Auto
- Incandescent
- Fluorescent
- Warm Fluorescent
- Daylight
- Cloudy Daylight
- Twilight

- Shade

By default, auto mode is enabled. You can select the required AWB modes from the **AWB mode** drop-down list box. When you click the **AWB mode** drop-down list box, the list of AWB modes appear as shown below.

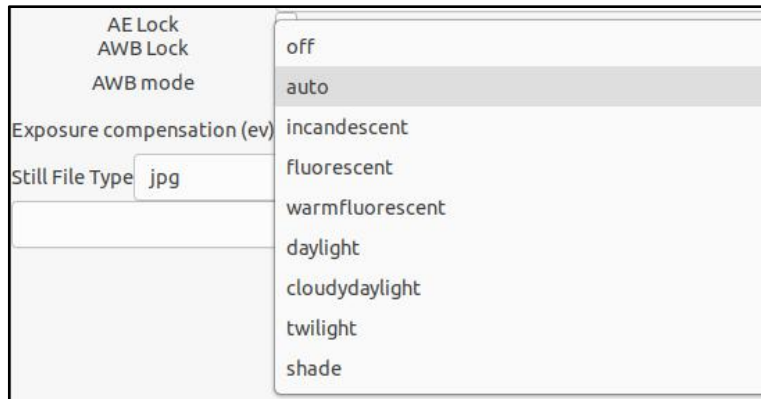


Figure 14: AWB Mode

Exposure Compensation

The exposure compensation control is used to set the threshold of the optimum exposed image. This control value can be changed from a minimum value of -2.00 ev to a maximum value of +2.00 ev. The default value for exposure compensation is 0.00 ev. You can change the exposure compensation value by manually inserting the values or selecting the value from the **Exposure compensation (ev)** drop-down list box. You can view the screen similar to the screen shown below.

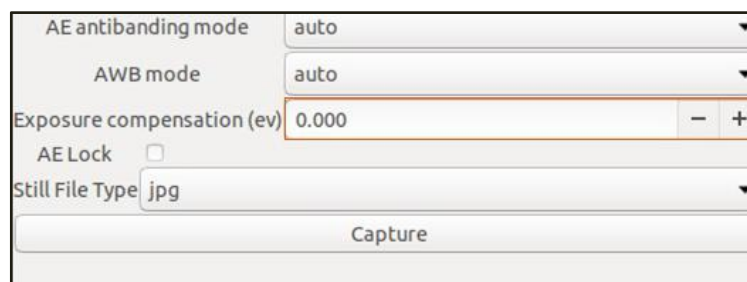


Figure 15: Exposure Compensation

AE Lock

The AE lock control is enabled during the capture and video module of eCAM_argus_camera. When you select the **AE Lock** check box, it locks the exposure to current values.

Still File Type

There are two types of still capture supported in the eCAM_Argus_Camera. You can select the type of still image capture to be JPEG compressed or uncompressed ISP processed YUV frame using the **Still File Type** drop-down list box provided in the capture module of eCAM_argus_camera.

When you click the **Still File Type** drop-down list box, you can view the supported still file type as shown below.

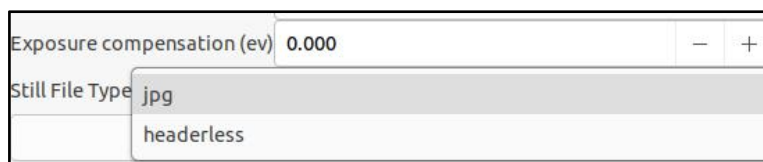


Figure 16: Still File Type

Video Recording

The eCAM_argus_camera also provides the feature for recording video.

When you click the **Capture** drop-down list box, you can view the option Video as shown below.

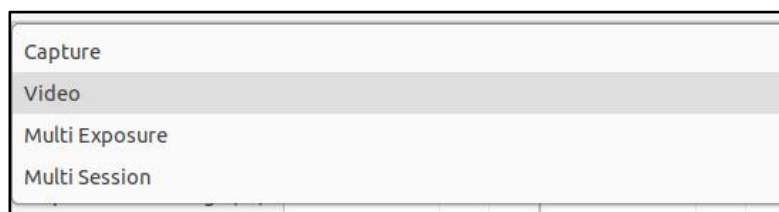


Figure 17: Video Mode

Click this video option to get the video recording controls.

Now, when you click the **Video Format** drop-down list box, the list of video recording formats will appear as shown below.

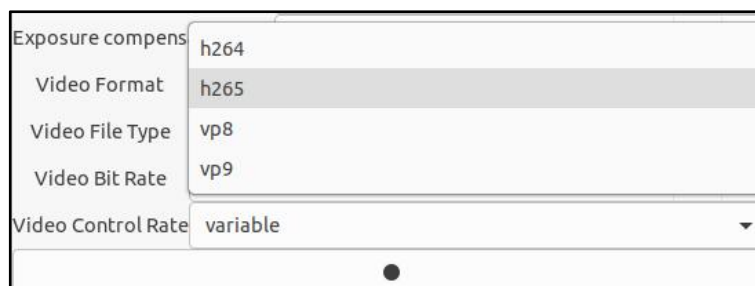


Figure 18: Video Recording Format

The video formats supported by the e-CAM121_CUONX for recording are as follows:

- H264
- H265

When you click the **Video File Type** drop-down list box, the list of supported file types will appear as shown below.

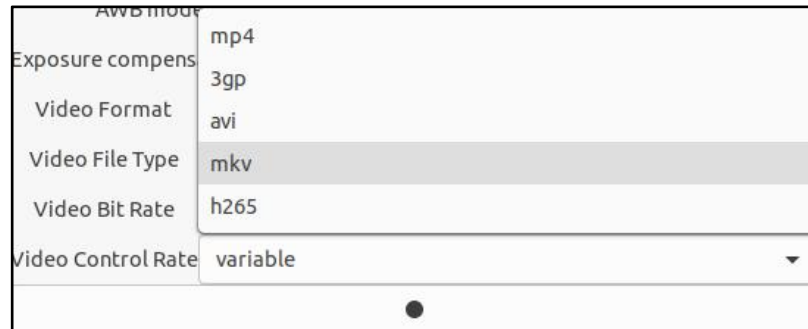


Figure 19: Video File Type

The supported video file types are as follows:

- MP4
- 3GP
- AVI
- MKV

Note:

If the resolution is **4K or higher**, the **H265** video encoding format must be used. In H265 video encoding format except **AVI** video file type, the rest three will support.

All the above-mentioned four video file types are supported in **H264** video encoding format. This format is supported only if the resolution is below 4K.

Note: To view the recorded videos, please ensure the GStreamer plugins are properly installed. Please refer to the *e-CAM121_CUONX_GStreamer_Usage_Guide_Rev_<ver>.pdf* to download the required plugins.

Frame Synchronization

e-CAM121_CUONX supports frame synchronization among cameras connected with the pulse provided externally.

The way to provide a pulse width modulation signal to our sensor, is listed below.

External Trigger

Here you must give the PWM trigger signal using external trigger signal connector (CN5) in the adaptor board. The CN5 connector position is shown in the below figure.

CN5 position(External Trigger header)

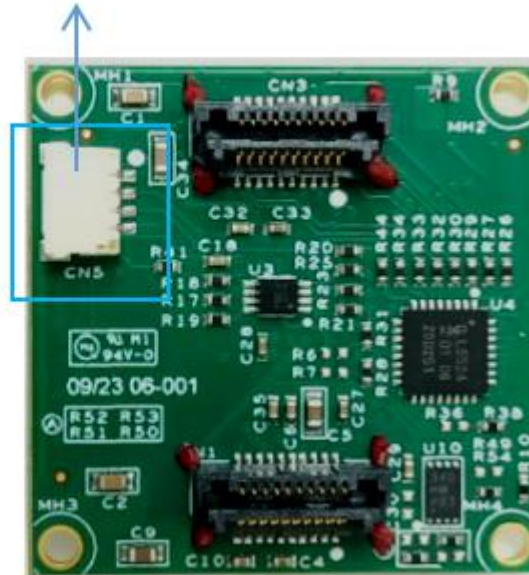


Figure 20: Connector for External Trigger Source

Pin description is listed in the below table:

CN5 Pin No	Signal Name	Pin Type	Description
1	VCC_3P3	POWER	3.3V Power supply from module
2	EXT_TRIGGER	INPUT	Camera Trigger signal
3	CAM_SHUTTER	OUTPUT	Shutter signal
4	GND	POWER	Ground signal for digital and analog

Note:

Please ensure to provide trigger signal of 3.3V or 5V for proper functionality. The voltage signal greater than 5V will cause permanent damage to the chip.

Enable sync:

Run the following command to enable Synchronization mode:

```
v4l2-ctl -d 0 -c frame_sync_mode=1
```

Disable sync:

Run the following command to disable synchronization mode.

```
V4l2-ctl -d 0 -c frame_sync_mode=0.
```

Launching Application

After enabling the sync mode, run the below commands in the host PC by connecting the development kit to host PC via ssh, to launch the application.

```
export DISPLAY=:0  
eCAM_argus_camera
```

Note: Make sure trigger pulse is already being generated before launching the application. Otherwise argus application will stop giving the stream which leads to [restarting nvargus-daemon](#) or restarting device at worst case.

Restart nvargus-daemon

Run the following commands to restart nvargus-daemon

```
sudo pkill nvargus-daemon  
sudo nvargus-daemon
```

Comparing eCAM_argus_camera with argus_camera by NVIDIA

This section describes the comparison of eCAM_argus_Camera with NVIDIA® argus_camera.

Users familiar with NVIDIA® Jetson Orin Nano™/Orin™ NX development kit may be familiar with NVIDIA® default camera application on Orin Nano™/Orin™ NX, argus_camera.

Additional features in NVIDIA® argus_camera and not in e-con Systems eCAM_argus_camera are as follows:

- Focus Position
- YUV Format
- Aperture motor step
- Aperture motor speed

Troubleshooting

In this section, you can view the list of commonly occurring issues and their troubleshooting steps.

When I capture JPEG image and leave the application on free running for one or two hours, I am getting a JPEG encoder timeout?

This is known limitation. Once the JPEG encoder timeout occurs, the streaming will be fine. But when you try to capture an image, the application may hang.

To recover the application, you must kill the application and run the following command.

```
$ sudo service nvargus-daemon restart
```

1. How can I get the updated package?

Please login to the [Developer Resources](#) website and download the latest release package.

What's Next?

After understanding the usage of eCAM_argus_camera application, you can refer to the following documents to understand more about e-CAM121_CUONX.

- *e-CAM121_CUONX Release Notes*
- *e-CAM121_CUONX Datasheet*

Glossary

CMOS: Complementary Metal Oxide Semiconductor.

ISP: Image Signal Processor.

MIPI: Mobile Industry Processor Interface.

L4T: Linux for Tegra.

Contact Us

If you need any support on e-CAM121_CUONX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

Revision History

Rev	Date	Description	Author
1.0	24-May-2023	Initial Draft	Product Marketing Team
1.1	01-July-2023	Support for both Orin Nano™ and Orin™ NX development kit	Product Marketing Team
1.2	27-August-2023	Supported Synchronous mode and 4K and 1080p resolution	Product Marketing Team